Iowa State University
Electrical And Computer Engineering

EE 2300

LAB 4 REPORT

October 1, 2025

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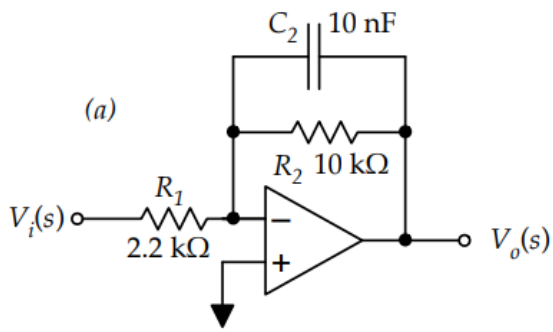
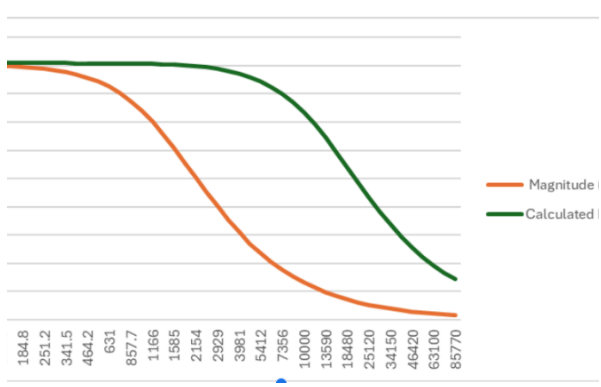
Introduction

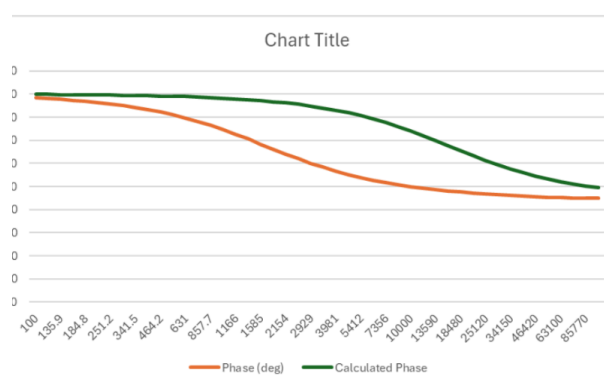
The purpose of this lab was to make characteristic observations on the behavior of second order filters. Through the design, building, and testing of Low-Q low-pass filters, various high-Q filters, and a Sallen-Key filter, it was possible to obtain a deeper understanding of the behavior of second order circuits.

A Low-Q low-pass active filters

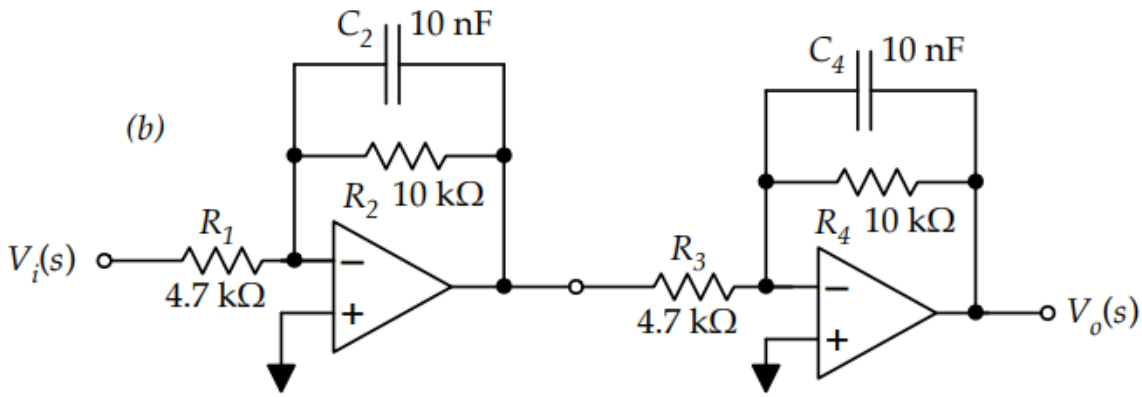


	Calculated	Measured
Corner Frequency, f_c	1591 Hz	1585 Hz
Gain, G_o	4.54	4.51

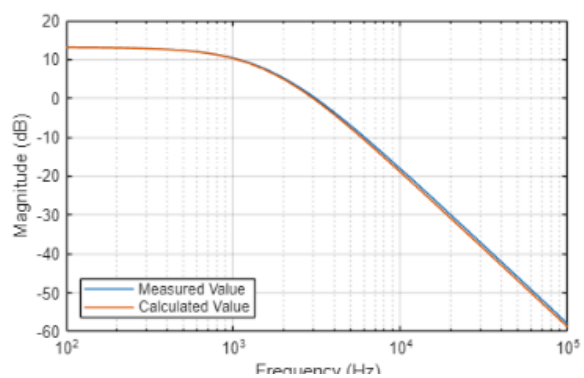


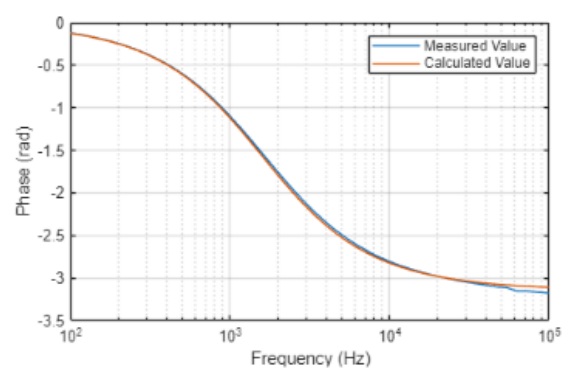


The measured corner frequency is in close agreement with the calculated value, differing by approximately 3%. This small deviation can be attributed to component tolerances in the resistors and capacitors, as well as minor deviations in the operational amplifier's behavior from the ideal model.



	Calculated	Measured
Corner Frequency, f_c	1.59 kHz	1556 Hz
Gain, G_o	2.127	2.1
Pole Quality Factor, Q_p	$10 * 10^{-4}$	0

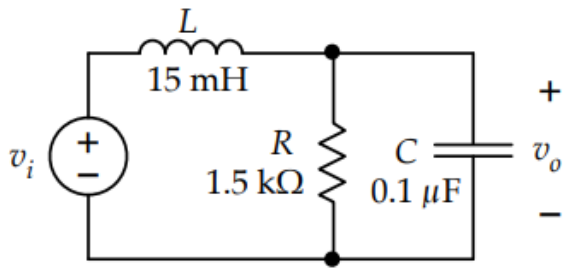




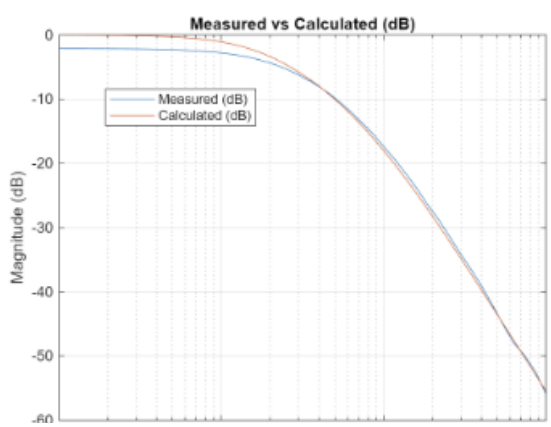
The second-order filter behaved consistently with the calculated outcome for measurement. Proper damping is occurring and there is a smooth roll-off of high frequencies occurring.

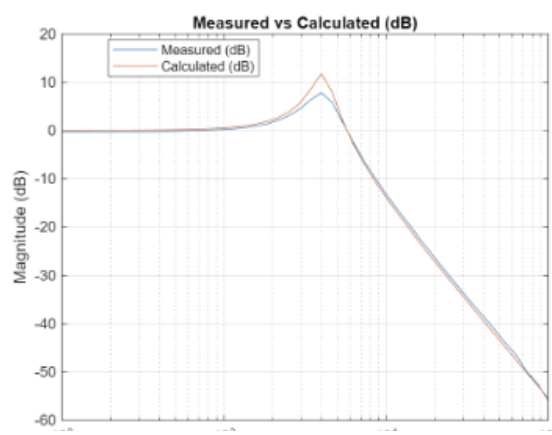
B RLC 2nd-order filters — high Q.

Figure 2. Low-pass.



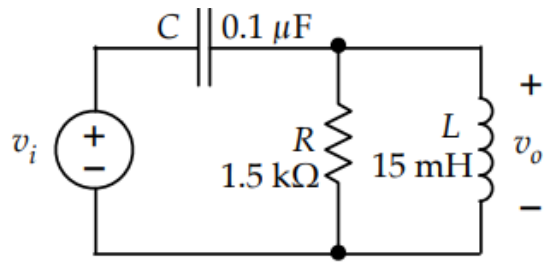
	Calculated	Measured
Corner Frequency, f_c	4,109 kHz	4062 Hz
Gain, G_o	$1.5 * 10^{-9}$	0
Pole Quality Factor, Q_p	3.873	3.704



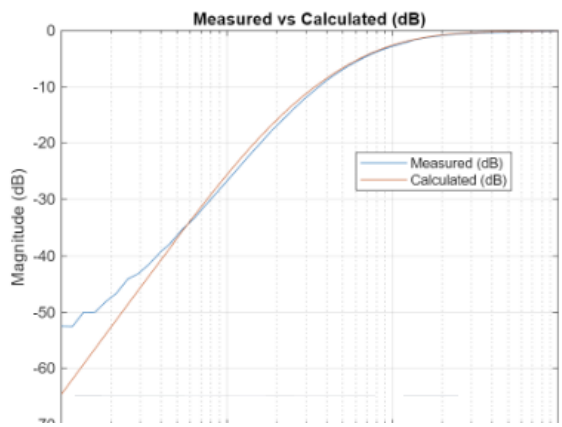


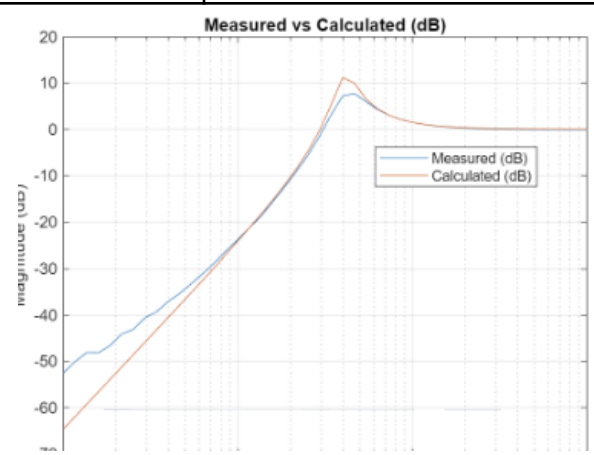
The parasitic resistance of the inductor likely caused the minor difference between the theoretical and measured outcomes for this filter. This would cause the Q factor to change and cause differing roll-off behavior than the ideal case.

Figure 3. High-pass



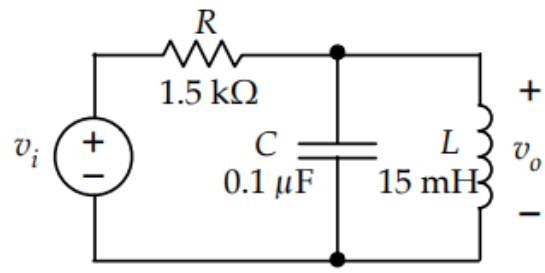
	Calculated	Measured
Corner Frequency, f_c	4,109 kHz	4000 Hz
Gain, G_o	$1.5 * 10^{-9}$	0
Pole Quality Factor, Q_p	3.873	3.510





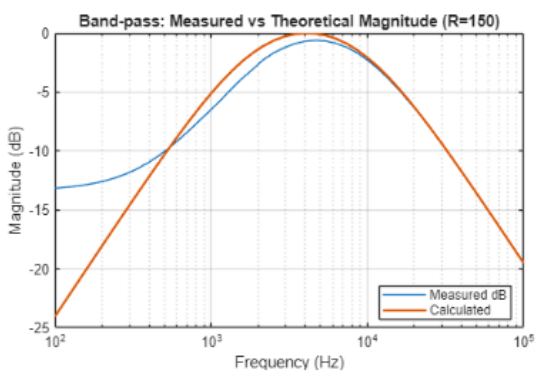
Differences between the theoretical and measured models may be attributed to the inductor and capacitor's losses due to internal resistance.

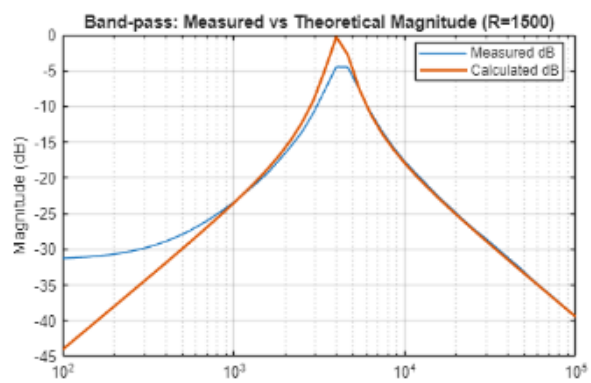
Figure 4. Band-pass



Complete the following table the band-pass circuit. Also provide plots containing both measured frequency responses.

	Calculated	Measured
Corner Frequency, f_c	4,109 kHz	3900 Hz
Gain, G_o	$1.5 * 10^{-9}$	0
Pole Quality Factor, Q_p	3.873	3.631

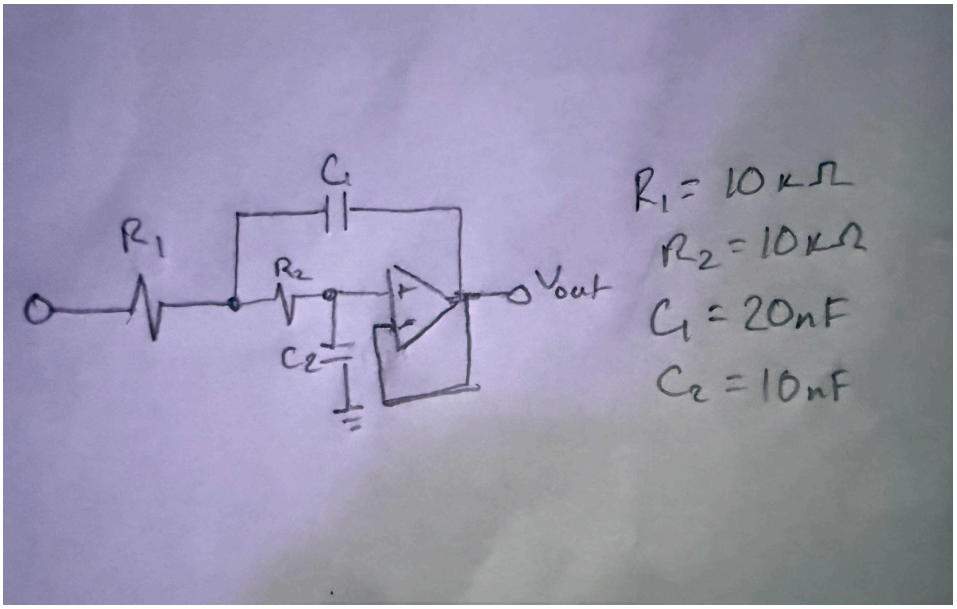


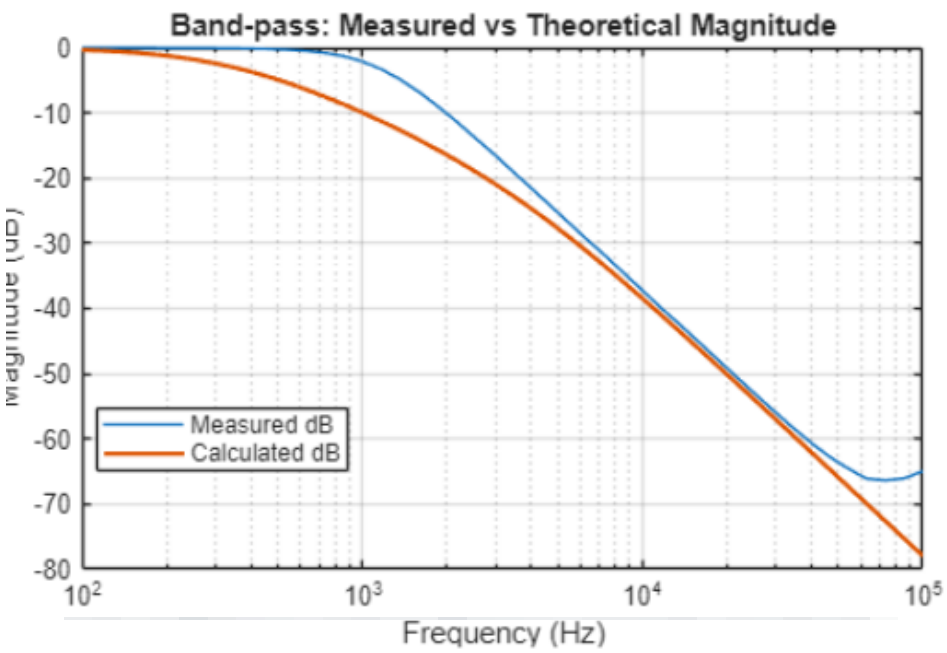


The band-pass circuit exhibits a distinct resonance near the predicted frequency. Minor differences between the calculated and measured resonant frequencies can be attributed mainly to the inductor's internal resistance and the capacitor's tolerance. These characteristics decrease the peak amplitude at resonance and broaden the frequency response, resulting in a lower Q factor.

C Active Filter Design

A Sallen-Key filter was designed to achieve a cutoff frequency of 1kHz and a quality factor of $Q = \frac{1}{\sqrt{2}}$. This resulted in the following circuit.



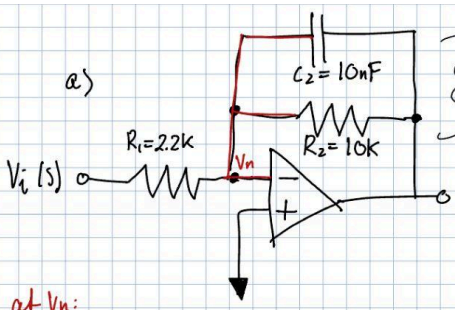


Conclusion

Through the design, building, and testing of Low-Q low-pass filters, various high-Q filters, and a Sallen-Key filter, it was possible to obtain a deeper understanding of the behavior of second order circuits. The comparison between calculated and measured values overall were in good agreement, but for those without good agreement, there were physical quantities such as parasitic resistance that played a role in the difference in outcomes.

Calculations

a)



Parallel callit Z_1: $\frac{\frac{1}{sC} \cdot R}{\frac{1}{sC} + R} =$

$$H(s) = \frac{b_2 s^2 + b_1 s + b_0}{a_2 s^2 + a_1 s^1 + a_0}$$

$$\omega_0 = \sqrt{\frac{a_0}{a_2}} = \qquad Q_p = \frac{\sqrt{a_0 a_2}}{a_1} = \frac{\omega_0 a_2}{a_1}$$

$$f_c = f_0 = \frac{\omega_0}{2\pi} = \omega_p$$

at V_n:

$$\frac{V_n - V_i}{R_1} + \frac{V_n - V_o}{Z_n} = 0 \qquad V_n \approx V_p = 0$$

$$\frac{-V_i}{R_1} + \frac{-V_o}{Z_n} = 0 \rightarrow \text{want } \frac{V_o}{V_i} \rightarrow \left(\frac{V_o}{Z_n} = \frac{-V_i}{R_1}\right) Z_n$$

$$10 \times 10^{-9}$$

0.000000010

$$V_o = \frac{-V_i Z_n}{R_1} \rightarrow \frac{V_i}{V_o} = \frac{-Z_n}{R_1} = \frac{\left(\frac{\left(-\frac{R_2}{sC}\right)}{\frac{1}{sC} + R_2}\right)}{R_1} = \frac{\frac{-R_2}{sC}}{R_1\left(\frac{1}{sC} + R_2\right)} = \frac{V_o}{V_i}$$

$$= \frac{-R_2}{sCR_1\left(\frac{1}{sC} + R_2\right)} = \frac{-R_2}{R_1 + sCR_1R_2} = \frac{V_o}{V_i} = H(s) \rightarrow \frac{b_0}{a_1 s^1 + a_0}$$

$$= \frac{-R_2/R_1}{1 + sCR_2} = H(s)$$

let $s = 0 \rightarrow g_0 = \frac{-R_2}{R_1} = \frac{10k}{2.2k} = \frac{10}{2.2} = \frac{10}{2 + \frac{1}{5}} = \frac{50}{10+1} = \frac{50}{11} = -4.\overline{54}\,\frac{\Omega}{\Omega} = g_0$

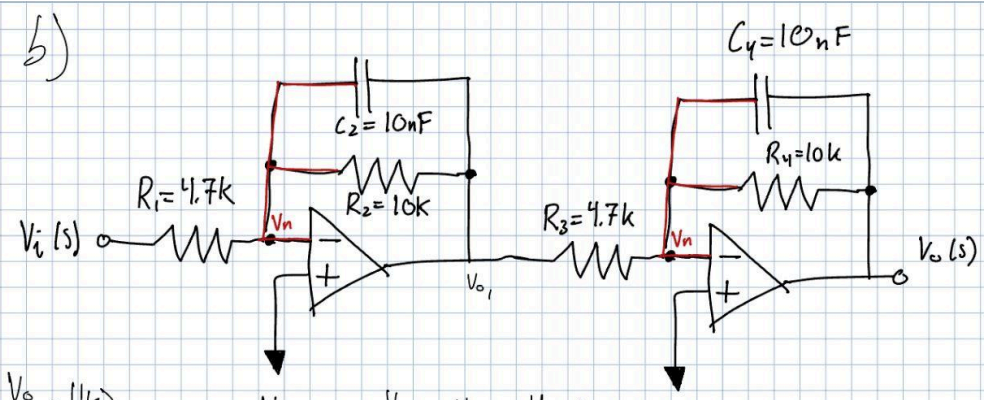
$= 4.\overline{54}\,\frac{\Omega}{\Omega} + \angle 180°$

$$\frac{\omega_p}{2\pi} = f_0 = f_c = \frac{1}{2\pi R_2 C} = \frac{1}{2\pi(10k)(10n)} = \frac{1}{2\pi(100\mu)} = \frac{1}{2\pi 10^{-4}} = 1{,}591.5\,Hz \approx 1.59\,kHz = f_0$$

$10^3 \cdot 10^{-9}$ $\quad 10^{-6}$

No Q_p → first order filter.

b)



$\frac{V_o}{V_i} = H(s)$ we want $\frac{V_{o2}}{V_{i1}}$ $\qquad \frac{V_o}{V_i} = H_1(s) \cdot H_2(s)$

$$\frac{-V_i}{R_1} + \frac{-V_{o1}}{Z_1} = 0 \rightarrow \frac{-V_i Z_1}{R_1} = V_{o1} \rightarrow \frac{-Z_1}{R_1} = \frac{V_{o1}}{V_i} \rightarrow \frac{\frac{R_2}{R_1}}{1+sCR_2} = H_1(s)$$

$H_2(s)$ will be the same

$H_1(s) \cdot H_2(s) = H^2(s)$

$$\frac{\left(-\frac{R_2}{R_1}\right)}{(1+sC_2R_2)} \cdot \frac{\left(-\frac{R_2}{R_1}\right)}{(1+sC_4R_4)} = \boxed{\frac{\left(\frac{R_2^2}{R_1^2}\right)}{1+2sCR_2+s^2C_{24}^2R_2^2} = H_{tot}(s)}$$

$R_2 = R_4$ $\quad R_1 = R_3$

$C_2 = C_4$

$$g(0) = \text{gain} = \frac{\left(\frac{(10k)^2}{(4.7k)^2}\right)}{1+0+0} = \frac{10k^2}{4.7k^2} = \frac{(10 \cdot \cancel{10^{3\cdot 2}})}{(4.7 \cdot \cancel{10^{3\cdot 2}})} = \frac{10}{4.7} = \boxed{2.127 = G_o}$$

$R_1 = 4.7k$
$R_2 = 10k$
$C_2 = 10n$
$\omega_p -$

$$f_c = f_o = \frac{\omega_o}{2\pi} = \frac{\omega_p}{2\pi} \rightarrow \frac{\left(\frac{1}{RC}\right)}{\left(\frac{2\pi}{1}\right)} = \frac{\left(\frac{1}{RC}\right)}{2\pi} = \frac{1}{(4.7k)(10n)(2\pi)} = \boxed{3.386kHz = f_o = f_c}$$

$$\sqrt{\frac{1}{R_1^2C^2}} = \frac{1}{R_1C}$$

$$\omega_o = \frac{1}{R_1C}$$

$$\left(\frac{\frac{R_2^2}{R_1^2}}{1+2sCR_2+s^2C^2R_2^2}\right)\left(\frac{\left(\frac{1}{C_{24}^2R_2^2}\right)}{\left(\frac{1}{C_{24}^2R_2^2}\right)}\right) = \frac{\frac{\cancel{R_2^2}}{R_1^2 \cdot C_{24}^2 \cancel{R_2^2}}}{\frac{1}{C^2R_2^2} + \frac{2s}{CR_2} + s^2} = \frac{\left(\frac{1}{R_1^2 \cdot C^2}\right)}{\left(s^2 + s\frac{2}{CR} + \frac{1}{C^2R_2^2}\right)}$$

Shouldn't be "2" (I think)
↳ should have kept C_2 & C_4 separate?
↳ algebraically it should be fine

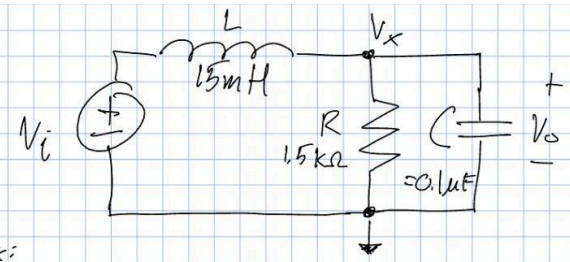
$$H(s) = G_o \cdot \frac{\omega_o^2}{s^2 + \left(\frac{\omega_o}{Q}\right)s + \omega_o^2}$$

$$Q = CR_2 = 10^{-4}$$

$Q = 10^{-4}$ $\qquad \omega_o = 21{,}276.6 \frac{rad}{s}$ $\qquad f_o = f_c = 3.386kHz$

$G_o = 2.127$

B) figure 2:



node voltage at V_x:

$$\frac{V_x - V_i}{sL} + \left[\frac{V_x - 0}{R} + \frac{V_x - 0}{C}\right] \Rightarrow \text{in parallel} \quad \left(\frac{R \cdot \frac{1}{sC}}{R + \frac{1}{sC}}\right)\left(\frac{sC}{sC}\right) = \frac{R}{sRC+1}$$

$$\frac{V_x - V_i}{sL} + \frac{V_x - 0}{Z_1} = 0 \quad \text{want } \frac{V_x}{V_i}$$

$$sL \cdot V_x \left(\frac{1}{sL} + \frac{1}{Z_1}\right) = V_i$$

$$\frac{V_x}{V_i} = \frac{1}{sL\left(\frac{1}{sL} + \frac{1}{Z_1}\right)} = \frac{1}{sL\left(\frac{1}{sL} + \frac{sCR+1}{R}\right)} = \frac{1}{1 + \frac{s^2CLR + sL}{R}} =$$

$$= \frac{1}{1 + s^2CL + \frac{sL}{R}} = \frac{1}{CLs^2 + \frac{sL}{R} + 1} = H(s) = \frac{1/LC}{s^2 + s \cdot \frac{1}{RC} + \frac{1}{LC}}$$

$$H(s) = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q} \cdot s + \omega_0^2} \qquad \omega_0^2 = \frac{1}{LC} \rightarrow \boxed{\omega_0 = \frac{1}{\sqrt{LC}}} \qquad \frac{\omega_0}{Q} = \frac{1}{RC} \rightarrow \frac{\left(\frac{1}{\sqrt{LC}}\right)}{Q} = \frac{1}{RC}$$

$$\left(\frac{RC}{\sqrt{LC}}\right)^2 = (Q)^2 \rightarrow \sqrt{\frac{R^2C^2}{LC}} = \sqrt{Q^2} \rightarrow \boxed{R \cdot \sqrt{\frac{C}{L}} = Q}$$

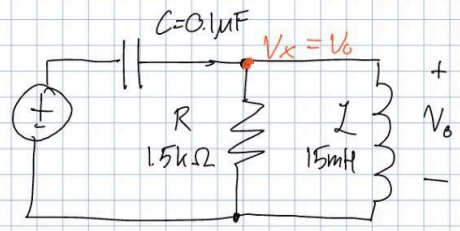
$$\frac{\omega_0}{2\pi} = \boxed{f_0 = \frac{1}{2\pi\sqrt{LC}}}$$

$R = 1.5\,k\Omega$, $L = 15\,mH$, $C = 0.1\,\mu F$

$\omega_0 = 25{,}820$ rad/s, $f_0 = 4{,}109$ Hz $\quad Q = 3.873$

$G_0 = 1.5 \times 10^{-9}$

B. figure 3.



$$R\|L = \frac{RsL}{R+sL} = Z_1$$

Voltage divider:

$$\frac{V_x}{V_i} = \frac{Z_1}{\frac{1}{sC}+Z_1} = \left(\frac{\left(\frac{RsL}{\cancel{R+sL}}\right)}{\frac{1}{sC}+\left(\frac{RsL}{\cancel{R+sL}}\right)}\right)\left(\frac{\cancel{R+sL}}{\cancel{R+sL}}\right) = \left(\frac{RsL}{\frac{R}{sC}+\frac{sL}{sC}+RsL}\right)\left(\frac{sC}{sC}\right) = \frac{RLCs^2}{R+sL+RLCs^2}$$

$$\frac{V_o}{V_i} = \frac{V_x}{V_i} = \frac{RLCs^2}{RLCs^2+sL+R}\left(\frac{\left(\frac{1}{R}\right)}{\left(\frac{1}{R}\right)}\right) = \frac{LCs^2}{LCs^2+\frac{sL}{R}+1} = H(s)$$

want "1"

Std form: $\dfrac{\left(\frac{G_0 s^2}{\omega_0^2}\right)}{\left(\frac{s^2}{\omega_0^2}\right)+\left(\frac{s}{Q\cdot\omega_0}\right)+1}$

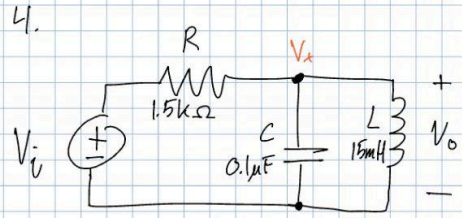
$\boxed{G_0 = LC}$

$$\frac{LC\cancel{s^2}}{1} = \frac{\cancel{s^2}}{\omega_0^2} \rightarrow \sqrt{\omega_0^2} = \sqrt{\frac{1}{LC}} \rightarrow \boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$

$$\frac{\omega_0}{2\pi} = \boxed{f_c = \frac{1}{2\pi\sqrt{LC}}}$$

$G_0 = 1.5\times10^{-9}$ $\quad \omega_0 = 25{,}820$ rad/s $\quad f_0 = 4{,}109$ Hz

B figure 4.



$$L\|C \rightarrow \frac{sL\cdot\frac{1}{sC}}{sL+\frac{1}{sC}} = \left(\frac{\frac{1}{C}}{sL+\frac{1}{sC}}\right)\left(\frac{sC}{sC}\right) = \frac{sL}{s^2LC+1}$$

Voltage divider: $$\frac{V_x}{V_i} = \frac{V_o}{V_i} = \frac{Z_1}{R+Z_1} = \left(\frac{\left(\frac{sL}{s^2CL+1}\right)}{R+\left(\frac{sL}{s^2CL+1}\right)}\right)\left(\frac{s^2CL+1}{s^2CL+1}\right) = \frac{sL}{s^2RCL+sL+R} = \frac{V_o}{V_i}$$

$$H_{BP}(s) = \frac{\frac{\omega_0}{Q}\cdot s}{s^2+\frac{\omega_0}{Q}\cdot s+\omega_0^2}$$

$$\cancel{s}L = \frac{\omega_0}{Q}\cdot\cancel{s} \qquad R = \omega_0^2$$

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad Q = R\sqrt{\frac{C}{L}}, \quad \frac{\omega_0}{Q} = \frac{1}{RC}$$

$Q = 3.873$ $\quad \omega_0 = 25{,}820$ $\quad f_0 = 4{,}109$ Hz

$G_0 = 1.5\times10^{-9}$